

1. Overview

Wreath product:

$$\text{Group } G \xrightarrow{\sim} G \wr \Sigma_d \hookrightarrow \Sigma_{md}$$

e.g. $C_2 \wr \Sigma_d =$ Weyl group of type B

$C_m \wr \Sigma_d = G(m, 1, d)$: Cplx refln grp

$\mathbb{Z} \wr \Sigma_d = \Sigma_d^{\text{ext}}$: affine sym grp

$\Sigma_m \wr \Sigma_d$: "less interesting" (NOT CRG)

B : assoc alg $\Rightarrow$ [Kleshchev-Muth '19] Aff'n alg

$B =$ zigzag alg $\leadsto$ ing strata of KLR alg

B : Frobenius alg $\Rightarrow$ [Rosso-Savage '20] Frob-Hecke alg

$B = K[X]G \leadsto$ [Wan-Wang '08] Wreath Hecke alg

B : comm. ring $\Rightarrow$ [Elias '22] Hecke-type coctries

Defn [Coi-Nakano-Xiang '23]

Quantum wreath product

$$\text{Assoc alg } B \xrightarrow{\sim} B \wr \mathbb{H} \xrightarrow{\sim} \mathbb{H}(\Sigma_m)$$

Sometimes

e.g. $B = \mathbb{H}(\Sigma_m) \Rightarrow$ Hecke alg $\mathbb{H}_{m,2} \hookrightarrow \mathbb{H}_q(\Sigma_m)$

(now) Hecke alg of $\Sigma_m \wr \Sigma_2 \leadsto \Sigma_m \wr \Sigma_d$

$B = K[X^{\pm 1}] \Rightarrow$ Affine Hecke alg $\mathbb{H}_q^{\text{Aff}}(\Sigma_d)$

Ariki-Koike alg $\mathbb{H}_{q,t}(\Sigma_m \wr \Sigma_d)$

$B = K[X] \Rightarrow$ deg AHA

$\Rightarrow$ Nil Hecke ring (HOM space of KLR alg)

Goal Determine nec (and suff.) conditions s.t.

struc theory of $B \Rightarrow$ struc theory of

$$A = B \wr \mathbb{H}$$

repr theory of $B \Rightarrow$ repr theory of A

Connections

1) 1-cover of Hecke alg $\mathbb{H}_q(G/P, P/d)$

2) Geom repr theory for $\Sigma_m \wr \Sigma_d$

3) Elias' generalized KLR alg

4) MacDonnell identity

2. Background

[Iwahori-Matsumoto '64-5]

(d.f.) Hecke algebra

$\hookrightarrow$ Chevalley group G / finite field

Fact

(a) G has a BN pair

i.e., (G, B, N, W) forms a Tits system:

① $G = \langle B, N \rangle$ ② $W \cong N/B \cap N = \langle s_i \rangle_{i \in I}$ s.t. $\forall s = s_i$

$$s^2 = 1, sBs \leq B \cup BwB, sBs \neq B \quad (w \in W)$$

(b) $\mathbb{H}(G, B)$ is a specialization of (generic) Hecke algebra (where $q \in \mathbb{C}^\times$):

$$\mathbb{H}_q(W) := \langle T_i \rangle_{i \in I} / T_i^2 = (q-1)T_i + q$$

braid relations

$\leadsto$ standard basis $\{T_w | w \in W\}$ where $T_w := T_{s_1} \cdots T_{s_n}$

for any rex $w = s_{i_1} \cdots s_{i_n}$

$$\leadsto \bar{T}_i = q^{-1}(T_i - (q-1))$$

$$\bar{q} := q^{-1}$$

$\leadsto$ bar involution $-$ on $\mathbb{H}_q(W)$ s.t. $\bar{\bar{T}}_i = T_i$

$$\bar{\bar{T}}_w = T_w^{-1}$$

$$\leadsto \text{Hecke alg } \mathbb{H}(G, B) \stackrel{\text{v.s.}}{=} \text{Span}_{\mathbb{C}} \{T_w | w \in W\} \subseteq \mathbb{C}G$$

$$\text{where } T_w := \frac{1}{\#B} \sum_{x \in BwB} x$$

standard basis

Moreover, W is a (finite) Coxeter group, i.e.,

$$W = \langle s_i | i \in I \rangle / s_i^2 = 1$$

braid (Dynkin diagram)

$\leadsto$ length function

$$\ell(w) := \# s_i \text{ in any reduced expression of } w$$

shortest among all expr

(c) $\mathbb{H}_q(W)$ is a (Frobenius) symmetric algebra, i.e.,

a fd assoc alg w/ nondeg assoc symmetric bilinear form

equiv, $\exists$ trace map $\text{tr}: \mathbb{H}_q(W) \rightarrow \mathbb{C}$ s.t. $\text{tr}(ab) = \text{tr}(ba)$

$$\text{In particular, } \text{tr}(\sum_w C_w T_w) = \mathbb{C}e$$

$$\beta(a, b) := \text{tr}(a^* b)$$

where $*$: $T_i \mapsto T_i$ is the anti-aut on $\mathbb{H}_q(W)$

$\leadsto \mathbb{H}_q(W)$ is self-injective

For $M \in \mathbb{H}_q(W)\text{-mod}$, M is proj $\iff M$ is inj

(d) $\mathcal{H}(W)$ has (uniq) Kazhdan-Lusztig basis

$$\{\tilde{C}^w\}_{w \in W} \text{ s.t. (let } v^2 = q)$$

$$\tilde{C}^w = C^w \in v^{-\text{deg}(w)} T_W + \sum_{x < w} v N[v] v^{-\text{deg}(x)} \tilde{C}^x$$

$\leadsto$ These coeff describe irreducible characters of Lie alg having Weyl group W

② If $n \geq d$ then double centralizer prop holds:

$$\text{End}_H(T) \cong S, \quad \text{End}_S(T) \cong H$$

$$\leadsto T \cong \bigoplus_{\lambda} L_{\lambda} \oplus S^{\lambda} \text{ as } (S-H)\text{-modules}$$

$L_{\lambda} \in S\text{-irr}, S^{\lambda} \in H\text{-irr}$

$\leadsto$ Schur functor

$$\mathcal{F} = \text{Hom}(T, -) : S\text{-mod} \rightarrow H\text{-mod}$$

$L_{\lambda} \mapsto S^{\lambda}$

③

KL basis of $\mathcal{H} \leadsto$ canonical basis of $\tilde{U}_n$

Want to study rep theory of $\mathcal{H}(W)$ when it's NOT semisimple

$\leadsto$ Find a quasi-hereditary cover of $\mathcal{H}(W)$
(for short, a $\mathcal{O}$ -cover)

idea: approximate numerical data of $\mathcal{H}(W)$ from those of a nicer algebra A

Fact

A is qh $\iff A\text{-mod}$ is a highest weight caty
due to [Cline-Parshall-Scott '88]

i.e. $\exists$ standard modules $\{\Delta_{\lambda}\}_{\lambda \in \Lambda}$ s.t.

$$(1) \text{End}(\Delta_{\lambda}) = \mathbb{C} \quad \forall \lambda \in \Lambda$$

$$(2) \text{Hom}(\Delta_{\lambda}, \Delta_{\mu}) \neq 0 \implies \lambda \leq \mu$$

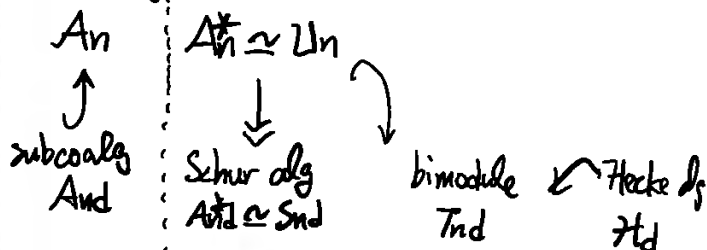
$$(3) \text{Hom}(\Delta_{\lambda}, M) = 0 \quad \forall \lambda \in \Lambda \implies M = 0$$

$$(4) \exists \text{ projective } P_{\lambda} \text{ s.t. } \ker(P_{\lambda} \rightarrow \Delta_{\lambda}) \text{ has a } \Delta\text{-filtration}$$

e.g. BGG caty $\mathcal{O}(g)$ for Lie alg

(e) Schur duality

Coord catg:



Idea: Study rep th simultaneously:

$$\mathcal{O} \text{ "deg } d \text{ repn" of } U_n \cong \text{And-conod} \cong S_n \text{ mod}$$

(f) $\mathcal{H}(W)$ is a cellular algebra, i.e., $\exists$ cell datum $(\Lambda, M, C, *)$ which gives

- (i) $\text{Irr } \mathcal{H}(W) = \{\text{non-vanishing cell modules}\}$
- (ii) criterion for (semisimplicity via decomp)

e.g. $W = \Sigma_n$ (type A) quasi-hereditary matrix

$$\Lambda = \{\lambda \vdash n\}, \quad M(\lambda) = \text{std}(\lambda)$$

$\leadsto$ cellular basis $\{C_{st} \mid s, t \in \bigcup_{\lambda \in \Lambda} M(\lambda)\}$ of $\mathcal{H}(\Sigma_n)$
(gives Robinson-Schensted algorithm)

Defn

A is a quasi-hereditary alg if $\exists$

chain $0 \subseteq I_1 \subseteq \dots \subseteq I_n = A$ of ideals s.t.

each $\mathcal{O}_j := I_j/I_{j-1}$ is a heredity ideal of A/I_{j-1}

$\leadsto A$ has finite global dimension

$$(i.e., \exists n \text{ s.t. } \text{Ext}_A^n(M, N) = 0 \quad \forall M, N \in A\text{-mod})$$

Thm [Dlab-Ringel '90]

Every fd alg A has a quasi-hereditary cover $\mathcal{O}$, i.e.

$\exists$ functor $\mathcal{F} = \text{Hom}(P, -) : \mathcal{O} \rightarrow A\text{-mod}$ s.t.

(1) $\mathcal{O}$ is a HWC ($\cong S\text{-mod}$ for $S: \mathcal{O}HA$)

(2) $P \in S\text{-proj}, A = \text{End}_S(P)$

(3) $\text{Hom}(M, N) \cong \text{Hom}_A(\mathcal{F}M, \mathcal{F}N) \quad \forall M, N \in S\text{-proj}$

e.g. Schur functor $S(n, d)\text{-mod} \rightarrow (\Sigma d\text{-mod})$
 $M \mapsto eM$

Thm [Rouquier '08]

the small and nice

Every fd alg A has a unique 1-cover, where

a qh cover is called an i -cover (i.e. $\mathcal{A}_i$) if
 $\text{Ext}_B^i(M, N) \cong \text{Ext}_A^i(\mathcal{A}_i M, \mathcal{A}_i N) \quad \forall M, N \text{ w/ } \Delta\text{-fils}$
 $j \leq i$

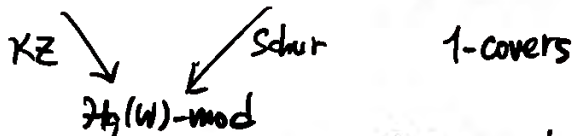
e.g. [Hammer-Nakano '04]

Schur functor $S_q(n, d) \rightarrow \mathcal{H}_q(\Sigma_d)\text{-mod}$ is a 1-cover if $\text{ord}(q) \geq 4$.

GGOR Problem

For $W: \text{CRG}$, $\begin{cases} \text{construct } S_q^W \\ \text{determine conditions s.t.} \end{cases}$

$$\mathcal{O}_c(HW) \xrightarrow{\sim} S_q^W\text{-mod as HWC}$$



Δ There're multiple non-isom algebras which can be called Schur alg of type B.
 Only one has 1-cover, to our knowledge

3. Hu Algebra $\mathcal{A}(m)$ [Hu'02]

Idea: $\mathcal{A}(m)$ is the "Hecke algebra" for $\Sigma_m \wr \Sigma_2$

$$\text{s.t. } \Sigma_m \times \Sigma_m \subseteq \Sigma_m \wr \Sigma_2 \subseteq \Sigma_m$$

quantifies to $\mathcal{H}_q(\Sigma_m \times \Sigma_m) \subseteq \mathcal{A}(m) \subseteq \mathcal{H}_q(\Sigma_m)$.

Recall $G^d \Sigma_d := G^d \times \Sigma_d \cong (g_1, \dots, g_d)W$
 $= W(g_{1w_0}, \dots, g_{dw_0})$

eg

$$G^2 \Sigma_d = \text{aut} \left(\begin{array}{c} \text{diagram with } d \text{ nodes and } 2 \text{ arrows} \end{array} \right), \quad \Sigma_m \wr \Sigma_2 = \text{aut} \left(\begin{array}{c} \text{diagram with } 2 \text{ nodes and } m \text{ arrows} \end{array} \right)$$

Consider $\mathcal{H}^B := \mathcal{H}_{(1,q)}(W(B_d)) \supseteq \mathcal{H} = \langle T_i | i > 0 \rangle$

braid $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \dots \begin{pmatrix} d-1 \\ 0 \end{pmatrix} \quad (i > 0)$

quad: $T_0^2 = 1, T_i^2 = (q-1)T_i + q$

Define Jucys-Murphy elements

$$U_0^\pm = 1, U_{i+1}^\pm := (T_i^\pm T_{i+1} T_i^\pm) U_i^\pm$$

$$\leadsto T_0 T_{i+1} U_i^\pm = T_{i+1} U_{i+1}^\pm - q^i T_{i+1} U_i^\pm$$

$$\leadsto T_0 \mathcal{H} U_i^\pm \subseteq \mathcal{H} U_i^\pm + \sum_{j \neq i} \mathcal{H} U_j^\pm \mathcal{H}$$

$\rightarrow$ Krizhnik-Zamolodchikov

e.g. [Ginzburg-Guay-Opdam-Rouquier '03]

KZ functor $\mathcal{O}_c(HW) \rightarrow \mathcal{H}_q(W)\text{-mod}$ is a qh-cover

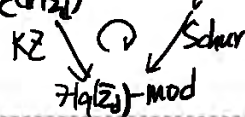
where $HW = \text{rational Cherednik alg}$

when $c \neq \frac{1}{2} + \sqrt{-1}$

$W = \text{any complex reflection gr}$

GGOR Conj [Rouquier '08]

$$\mathcal{O}_c(H_{\Sigma_d}) \xrightarrow{\sim} S_q^W\text{-mod as HWC}$$



when $c \neq \frac{1}{2} + \sqrt{-1}$
 $q^2 \neq -1$

GGOR(B) solved in [LNX'22]

Construction of 1-cover inspired by

$$\mathcal{H}_q(W(B_d)) \cong_{\text{Mor}} \prod_i \mathcal{H}_q(\Sigma_i \times \Sigma_{d-i})$$

GGOR($G(P, d)$) e.g. $W(D_d) = G(2, 2, d)$

Needs to study analogous Morita equiv:

$$\mathcal{H}_q(W(D_d)) \cong_{\text{Mor}} \begin{cases} \prod_i \mathcal{H}_q(\Sigma_i \times \Sigma_{d-i}) & \text{if } d = 2m+1 \\ \mathcal{A}(m) \oplus \prod_i \mathcal{H}_q(\Sigma_i \times \Sigma_{d-i}) & \text{if } d = 2m \end{cases}$$

The generator $t = \begin{array}{c} 1 \dots m \quad m+1 \dots 2m \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \end{array}$ has rex

$$t = S_m \rightarrow 2m+1 S_{m-1} \rightarrow 2m-2 \dots S_1 \rightarrow m, \ell(t) = m^2$$

Δ While $\Sigma_m \wr \Sigma_2 = \langle \Sigma_m \times \Sigma_m, t \rangle$,

$\langle \mathcal{H}_q(\Sigma_m \times \Sigma_m), T_t \rangle$ has $\dim > \# \Sigma_m \wr \Sigma_2$

b/c T_t has a minimal poly of deg > 2 .

Prop/Defn [Hu'02] Let $u^\pm = U_m^\pm$.

(a) $\exists!$ elt $h_m \in \mathcal{H}$ st.

$$u^- T_t u^+ \in h_m u^+ + \sum_{j \neq m} \mathcal{H} U_j^\pm \mathcal{H}$$

(b) $\exists!$ elt $z_m \in \mathbb{Z}(\mathcal{H}_q(\Sigma_m \times \Sigma_m))$ st.

$$u^- T_t u^+ T_t u^- T_t u^+ = z_m u^- T_t u^+$$

$$(c) h_m^2 = z_m$$

Defn The Hu alg is the subalg

$$A(m) := \langle H_q(\Sigma_m \times \Sigma_m), h_m \rangle \subseteq H_q(\Sigma_{2m})$$

Prop [Hu'02]

(a) $A(m)$ has a basis $\{T_w, T_w h_m \mid w \in \Sigma_m \times \Sigma_m\}$

(b) $A(m)|_{q=1} = \mathbb{C} \Sigma_m \Sigma_m$
 $h_m|_{q=1} = z^m t$ B = $H_q(\Sigma_m)$

(c) $A(m) \cong \langle B \otimes B, H_1 \rangle / (a \otimes b) H_1 = H_1 (b \otimes a)$
 $H_1^2 = \Sigma_m$
 $\Sigma_m \in \mathbb{Z}(B \otimes B)$

This construction can produce a subalg

$$H_q(mid) := \langle H_q(\Sigma_m^d), h_m, \dots, \pi_m^{m-1}(h_m) \rangle \subseteq H_q(\Sigma_{md})$$

where $\pi_m: T_i \mapsto T_{i+m}$

Δ can't obtain presentation (unable to check quad rel's for w/o explicit formula for h_m)

New formula (nicer when normalized)

Recall $T_i^2 = (q-1)T_i + q \quad (i > 0)$

$$\Rightarrow D = (T_i - q)(T_i + 1) = (T_i - v^2)(T_i + 1)$$

$$= (\frac{T_i}{v} - v)(\frac{T_i}{v} + v')$$

$\Rightarrow$ normalized std basis $\{I_i := \frac{T_i}{v}, I_0 := T_0\}$
 with quad rel'n $I_i^2 = (v - v')I_i + 1$
 $\leadsto I_i^{-1} = I_i + (v' - v)$

e.g. $m=1$

$$h_m = I_1 + I_1^{-1} = 2I_1 + (v' - v)$$

$$\leadsto z_m = h_m^2 = 4(I_1^2 + (v' - v)I_1) + (v' - v)^2 = (v' + v)^2$$

e.g. $m=2$ ($I_2 I_2 I_1 I_2$)

$$h_m = I_{2312} + I_{23T_2} + (I_{2312} + I_{25T_2}) I_3^2$$

While $I_t = I_{2312}$ has complicated powers,
 $h_m^2 \in \langle I_1, I_3 \rangle$ and is central!

Let $w_{m,i} = \overset{1 \dots m}{\times} \overset{m+1 \dots m+i}{\times}$ so $t = w_{m,m}$

Determine the uniq elt $h_{m,i}$ s.t.

$$u_i^{-1} T_{w_{m,i}} u_i^+ \in h_{m,i} u_m^+ + \sum_{j>m} H u_j^+ H$$

Lem $h_{m,1} = v^m (I_{1 \rightarrow m} + I_{\overline{1 \rightarrow m}})$

$$h_{m,i+1} = v^{m+2i} \left(\begin{aligned} &I_{i+1 \rightarrow i+m} h_{m,i} \\ &+ I_{i+1 \rightarrow i+m} h_{m,i} I_{m+1 \rightarrow m+i} I_{m+1 \rightarrow m+i} \end{aligned} \right)$$

Defn Let $I_{a \rightarrow b} := I_{a \rightarrow b}, I_{\overline{a \rightarrow b}} = \overline{I_{a \rightarrow b}} = (I_{b \rightarrow a})'$

Defn For $\varepsilon = (\varepsilon_1, \dots, \varepsilon_m) \in \{+, -\}^m$, set

$$I_\varepsilon := I_{m \rightarrow 2m-1}^{\varepsilon_1} \dots I_{1 \rightarrow m}^{\varepsilon_m}$$

$$C_\varepsilon := \prod_{i: \varepsilon_i = -} C_{m-i}, \text{ where } C_{i+1} := I_i C_i I_i$$

$C_0 = 1, C_1 = I_m^2$

Thm [LWX '22]

The extra generator $h_m \in A(m)$ is given by

$$h_m = v^{m(2m-1)} \sum_{\varepsilon \in \{+, -\}^m} I_\varepsilon C_\varepsilon$$

Bar-Invariant Basis

h_m behaves nicely wrt bar-involution

$$\leadsto b_t := v^{-m(2m-1)} h_m \overline{I_{w_0(\Sigma_m)}}$$

$\leadsto$ bar-invariant basis $\{b_w \mid w \in \Sigma_m \Sigma_m\}$
 s.t. $\overline{b}_w = b_w \in I_w + \sum_{y < w} v N[w] I_y$

Δ $b_x b_y \notin \sum_{z} N[v^{\pm 1}] b_z$, unlike CB

Conj b_t has a positive expansion in terms of dual KL basis, i.e.,

Let $\{C_w \mid w \in \Sigma_{2m}\}$ be the uniq basis of $H_q(\Sigma_{2m})$
 s.t. $\overline{C}_w = C_w \in I_w + \sum_{y < w} (-v')/N[-v'] I_y$
 (which satisfies $C_x C_y \in \sum_z N[-v^{\pm 1}] C_z$)

Then

$$b_t \in \sum_{z \leq \Sigma_{2m}} N[v^{\pm 1}] C_z$$

NCTS Lecture Series on QWP Day 2

Last time

$$H_u \text{ alg } A(m) := \langle H_q(\Sigma_m \times \Sigma_m), h_m \rangle \in H_q(\Sigma_{2m})$$

$$\downarrow$$

$$H_q(m, 2) := \langle H_q(\Sigma_m)^{\otimes 2}, H_1 \rangle / H_1^2 = \Sigma_m$$

$$H_1(a \otimes b) = (b \otimes a) H_1$$

Today

Axiomatized treatment that also deals w/ variants of Hecke algebras

1. Defining QWP

Defn [Lai-Nakano-Xiang '23]

Let B : assoc. alg / comm. ring K .

$$Q = (R, S, \Delta, \rho) \in \mathbb{Z}(B \otimes B)^2 \times \text{End}(B \otimes B)^2$$

$$\leadsto X_i := 1^{\otimes i-1} \otimes X \otimes 1^{\otimes d-i-1} \in B^{\otimes d} \quad (X = R \text{ or } S)$$

$$\psi_i: b_1 \otimes \dots \otimes b_d \mapsto \dots \otimes \psi(b_i \otimes b_{i+1}) \otimes \dots \in \text{End}(B^{\otimes d})$$

$$\leadsto B\mathbb{Z}H_d = B\mathbb{Z}H_d = \langle B^{\otimes d}, H_i \rangle$$

$$\begin{matrix} \text{braid } (H_i \dots H_{i+1}) \\ H_i^2 = S_i H_i + R_i \\ H_i b = \Delta_i(b) H_i + \rho_i(b) \end{matrix}$$

Thm B Let B be a sym Frob alg.

$$\begin{matrix} \text{Splitting of } T \\ \text{for } B \end{matrix} \xrightarrow{\text{certain assumptions}} \begin{matrix} \text{Splitting of } T \\ \text{for } A := B\mathbb{Z}H_d \end{matrix}$$

$$\Downarrow \text{Schur duality for } T_B \curvearrowright B \quad \Downarrow \text{Frobenius} \quad \Downarrow \text{Schur duality for } T_A \curvearrowright A$$

B is a normal subalg of A if $AB^+ = B^+A$

$$\leadsto - \otimes_B A \text{ and } A \otimes_B - \text{ are exact}$$

Applying them on $0 \rightarrow B^+ \rightarrow B \xrightarrow{\varepsilon} K \rightarrow 0$, we get SES

$$\begin{matrix} 0 \rightarrow B^+A \rightarrow A \rightarrow K \otimes_B A \rightarrow 0 \\ 0 \rightarrow AB^+ \rightarrow A \rightarrow A \otimes_B K \rightarrow 0 \end{matrix}$$

i.e. quotient alg

$$A//B := A/B^+A = A \otimes_B K \subseteq K \otimes_B A$$

Wreath product

$$G \wr \Sigma_d = G^d \rtimes \Sigma_d \xrightarrow{\quad} B\mathbb{Z}H_d = \langle B^{\otimes d}, H_1, \dots, H_{d-1} \rangle$$

$$(g, h) s = s(h, g) \quad \text{Wreath relations} \quad H(a \otimes b) = \Delta(a \otimes b) H + \rho(a \otimes b)$$

$$s^2 = 1 \quad \text{quadratic relations} \quad H^2 = S H + R$$

$$s_i s_j \dots = s_j s_i \dots \quad \text{braid relations} \quad (s, r \in \mathbb{Z}(B \otimes B))$$

$$H_i H_j \dots = H_j H_i \dots$$

Q: When does $B\mathbb{Z}H_d$ have basis of right size?

Thm A Let $\{b_i | i \in I\}$ be a basis of B .

$$H_w := H_{i_1} \dots H_{i_N} \text{ for } w \in \Sigma_d, w = s_{i_1} \dots s_{i_N}$$

then $\{b_{i_1} \otimes \dots \otimes b_{i_d} H_w | i_j \in I, w \in \Sigma_d\}$ - only a basis of $B\mathbb{Z}H_d \iff$ (A*) hold

! Defn of $B\mathbb{Z}H_d$ carries a regular repn $\leadsto$ Humphreys-type proof of basis thm

Rank $B\mathbb{Z}H_d$ does not always contain a subalg that's $\cong H_q(\Sigma_d)$.

In contrast, $B\mathbb{Z}H_d$ contains a Hecke alg as a "quotient" as long as B has a nice augmented algebra structure.

Defn [Ginzburg-Kumar '93]

An augmented alg is a K -alg B w/ counit $\varepsilon: B \rightarrow K$

$\leadsto$ augmentation ideal $B^+ := \ker \varepsilon \trianglelefteq B$

Prop Assume that (A*) hold. B is augmented.

If $\Delta(a \otimes b) \in K(b \otimes a)$, $\varepsilon(\rho(B^+)) = 0$

then $B^{\otimes d} \trianglelefteq A := B\mathbb{Z}H_d$ and

$$A//B \cong H_q(\Sigma_d) := \langle T_1, \dots, T_{d-1} \rangle$$

$$\text{braid } (T_i \dots T_{i+1})$$

(PF) It suffices to check, $T_i^2 = \varepsilon(s_i) T_i + \varepsilon(R)$

for $b \in \ker(\varepsilon)$, $H b = \Delta_i(b) H + \rho_i(b) \in (\ker \varepsilon) A$

2. Necessary condns

Idea: basis $\{b_n H^n\}$ exists

$\Rightarrow \exists$ No ambiguities for the following expressions;

2-strand relns (i.e., for $a, b \in B \otimes B$)

Hab



H^2a



H^3



H^3 has no ambiguity

$$\Leftrightarrow \begin{cases} p(S) = (S - \Delta(S))S + (R - \Delta(R)) \\ p(R) = (S - \Delta(S))R \end{cases}$$

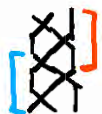
H^2a has no ambiguity

$$\Leftrightarrow \begin{cases} r_S \Delta = r_S \Delta^2 + p \Delta + \Delta p \in \text{End}(B \otimes B) \\ r_R = r_R \Delta^2 + p^2 - r_S p \end{cases}$$

where r_X = right multn by $X \in B \otimes B$

Will prove: (★1-5) are all conditions we need while there are others (which are conseq of above)

e.g. $H_{12121} \Leftrightarrow R_1 p_2(R_1) = 0 + \dots$



e.g. $(Ha)b \stackrel{?}{=} H(ab)$

$$\text{LHS} = \Delta(a)(Hb) + p(a)b$$

$$= \Delta(a)\Delta(b)H + \Delta(a)p(b) + p(a)b$$

$$\text{RHS} = \Delta(ab)H + p(ab)$$

$$\Rightarrow \text{requires } \begin{cases} \Delta(ab) = \Delta(a)\Delta(b) \\ p(ab) = \Delta(a)p(b) + p(a)b \end{cases} \quad (\star 1)$$

i.e., p is a Δ -twisted derivation

3-strand relations: (for $a \in B^{\otimes 3}$)

$H_{121}a$



H_{1211} (i.e. check $H_{12}H_1^2 = H_1^2H_{12}$)



$$\Delta_{121} = \Delta_{212}$$

$$\begin{aligned} &\vdots \\ &p_{21} + r_{R_1} \Delta_1 p_{21} \\ &= p_{212} + r_{R_2} \Delta_2 p_{21} \end{aligned}$$

$$\begin{aligned} &X_1 = \Delta_1(X_2), \quad p_2 \Delta_1(X_2) = 0 \\ &\quad X_2 \in \{S, R\} \\ &\Delta_2 p_1(S_2)S_2 + p_2 p_1(S_2) + \Delta_2 p_1(R_2) = 0 \\ &p_2(R_2) + \Delta_2 p_1(S_2)R_2 = 0 \end{aligned}$$

But if we assume

$$\Delta = \text{flip}, \quad p = 0,$$

then (★1-5) simplify to

$$\begin{cases} R = \Delta(R), \\ (\Delta(S) - S)R = 0, \\ r_S(\text{id} - \Delta) = 0. \end{cases}$$

3. Examples of QWP Algebras

3.1 Hu Alg

Set $B = \mathcal{H}_q(\Sigma_m)$,

$$\begin{cases} H^2 = \Sigma_m \\ H(a \otimes b) = (b \otimes a)H \end{cases} \Rightarrow \begin{cases} S = 0, \quad R = \Sigma_m \\ \Delta = \text{flip}, \quad p = 0 \end{cases}$$

Only need to verify $\Sigma_m = \Delta(\Sigma_m)$ (skipped)

3.2. AHA

Set $B = K[X^{\pm 1}]$.

$$\begin{cases} H^2 = (q-1)(1 \otimes 1)H + q(1 \otimes 1) \end{cases}$$

$$\begin{cases} H(X \otimes 1) = (1 \otimes X)H - (q-1)1 \otimes X \end{cases}$$

$$\Rightarrow \begin{cases} S = (q-1)1 \otimes 1, \quad R = q(1 \otimes 1) \end{cases}$$

$$\begin{cases} \Delta = \text{flip}, \quad \Delta: X \otimes 1 \mapsto -(q-1)1 \otimes X \end{cases}$$

Claim: Δ is uniquely determined by $\Delta(X \otimes 1)$ as long as p is a Δ -twisted derivation

Lem $p(x^i \otimes x^j) = \partial(x^i \otimes x^j) \overset{\text{call it } \beta}{p(x \otimes 1)}$

where $\partial(a \otimes b) := \frac{a \otimes b - b \otimes a}{x \otimes 1 - 1 \otimes x}$ is the Demazure op.
(pf) suffices to prove it for $i, j \geq 0$. By induction,

$$\begin{aligned} p(x^i \otimes 1) &= p(x^{i-1} \otimes 1)(x \otimes 1) \\ &= \partial(x^{i-1} \otimes 1) \beta + p(x^{i-1} \otimes 1)(x \otimes 1) \\ &= (1 \otimes x^{i-1}) \beta + (x \otimes 1) p(x^{i-1} \otimes 1) \end{aligned}$$

Hence,

$$\begin{aligned} p(1 \otimes x^j) &= p(1 \otimes x)(1 \otimes x^{j-1}) \\ &= (x \otimes 1) p(1 \otimes x^{j-1}) - \beta (1 \otimes x^{j-1}) \\ &= \dots = \beta (-1 \otimes x^{j-1} - x \otimes x^{j-2} - \dots) \\ &= \partial(1 \otimes x^j) \beta \end{aligned}$$

This lem makes verification of $(\star)$ much simpler.

Ariki-Koike Algebra

$$\mathcal{H}_{q, \vec{q}}(\text{Cm} \Sigma_d) := \frac{\langle Y, T_1, \dots, T_{d-1} \rangle}{\text{braid}(T_1 \dots T_{d-1})}$$

$$T_i^2 = (q-1)T_i + q$$

$$\prod_{i=1}^m (Y - q_i) = 0$$

$$Y T_i Y T_i = T_i Y T_i Y$$

$$Y^{(1)} Y^{(2)} = Y \otimes Y = Y^{(2)} Y^{(1)}$$
 is well-defined

Degenerate Artin $\{ \text{Nil Hecke alg} \}$

$$\mathcal{H}^{\text{deg}}(\Sigma_d) = \frac{\langle X_i, S_j \mid 1 \leq i \leq d, 1 \leq j \leq d-1 \rangle}{\text{braid}(S_1 \dots S_{d-1})}$$

$$\text{NH}(\Sigma_d) \quad \partial_i^2 = 0$$

$$S_i^2 = 1, \quad X_i X_k = X_k X_i$$

$$S_i X_i = X_{i+1} S_i - 1, \quad S_i X_k = X_k S_i$$

$$S_i X_{i+1} = X_i S_i + 1$$

Fact
 $\mathcal{H}^{\text{deg}}(\Sigma_d) \cong K[X] \wr \mathcal{H}_d$ with $\begin{cases} S_i \mapsto T_i \\ X_i \mapsto X^{(i)} \end{cases} \quad \begin{cases} S=0, R=1 \\ \delta = \text{flip} \\ \rho = -\partial \end{cases}$

$$\begin{aligned} &= \dots = (1 \otimes x^{i-1} + x \otimes x^{i-2} + \dots + x^{i-1} \otimes 1) \beta \\ &= \partial(x^i \otimes 1) \beta \end{aligned}$$

Next,

$$\begin{aligned} p(x \otimes x) &= p((x \otimes 1)(1 \otimes x)) = p((1 \otimes x)(x \otimes 1)) \\ &\stackrel{\parallel}{=} (1 \otimes x)(p(1 \otimes x) + \beta) \stackrel{\parallel}{=} (x \otimes 1)(p(1 \otimes x) + \beta) \\ &\Rightarrow p(1 \otimes x) = -\beta. \end{aligned}$$

Affine Hecke alg (of type A)

$$\mathcal{H}_q^{\text{ext}}(\Sigma_d) := \frac{\langle Y^\lambda, T_i \mid \lambda \in \mathbb{Z}^d, 1 \leq i \leq d-1 \rangle}{\text{braid}(T_1 \dots T_{d-1})}$$

$$T_i^2 = (q-1)T_i + q, \quad Y^\lambda \mu = Y^{\lambda+\mu}$$

$$T_i Y^\lambda = Y^{S_i(\lambda)} T_i + (q-1) \frac{Y^\lambda - Y^{S_i(\lambda)}}{1 - Y^{\alpha_i}}$$

Fact

$$\mathcal{H}_q^{\text{ext}}(\Sigma_d) \cong K[X^{\pm 1}] \wr \mathcal{H}_d, \quad T_i \mapsto H_i, \quad Y^\lambda \mapsto X^\lambda \otimes \dots \otimes X^\lambda$$

Fact

$$\mathcal{H}_{q, \vec{q}}(\text{Cm} \Sigma_d) \cong \frac{K[X^{\pm 1}] \wr \mathcal{H}_d}{\langle \prod_{i=1}^m (X - q_i) \otimes 1 \rangle^{\otimes d-1}}$$

$$\begin{aligned} T_i &\mapsto H_i \\ Y &\mapsto X^{(1)} \end{aligned}$$

$$\Delta \quad \mathcal{H}_{q, \vec{q}}(\text{Cm} \Sigma_d) \not\cong \left(\frac{K[X^{\pm 1}]}{\prod_i (X - q_i)} \right) \wr \mathcal{H}_d$$

b/c $T_i Y T_i$ doesn't satisfy the cyclotomic equation.

Affine Zigzag Algebra $\mathbb{Z}^{\text{aff}}(\Gamma)$

Γ = Dynkin diag of type ADE $\rightsquigarrow \bar{\Gamma}$: double quiver

$$B = \mathbb{Z}_{K[X]}^{\bar{\Gamma}} := K[X] \bar{\Gamma} / \text{path}(\partial \geq 3) = 0$$

$$\text{path}(\partial = 2) = 0 \text{ unless its cycle}$$

$$\text{cycles at same node are equal}$$

$$\text{v.s. } \{ e_i, a_{ij}, c_i \mid i \neq j \}$$

$S=0, R=1, \delta = \text{flip}, \rho$ is determined by

$$x_i \otimes x_j \mapsto 0, \quad x_i \otimes x_j = \begin{cases} (c_i + c_j) e_i \otimes e_j & \text{if } i=j \\ a_{ji} \otimes a_{ij} & \text{otw.} \end{cases}$$

Fact $\mathbb{Z}^{\text{aff}}(\Gamma)$

$$\mathbb{Z}_d(\Gamma) \cong \mathbb{Z}_{K[X]}^{\bar{\Gamma}} \wr \mathcal{H}_d$$

4. Basis Theorem

Now we show $(\star)$ holds $\Rightarrow \exists$ basis.

Recall Humphreys' proof of basis thm for $\mathcal{H}(W)$

Step 1. Consider free K -mod $\mathcal{E} = \bigoplus K T_w$

Step 2. Show $\mathcal{E} \cong \mathcal{H}(W)$ as $\pi \in \mathcal{H}(W)$ -mod using right multn $f_s: \mathcal{E} \rightarrow \mathcal{E}$
 $T_w \mapsto T_w s = \begin{cases} T_{ws} \\ (q-1)T_w + qT_{ws} \end{cases}$

i.e. for $w \in \Sigma_d$, assign a rex $r(w)$

s.t. $r(ws_i) = r(w)s_i$

$\leadsto f_a = f_a(r)$ is well-defined

(f_i is determined once f_a is)

We need to prove by induction on ℓ from the filtration $(\mathcal{E}^\ell)_\ell$ of $\mathcal{E}$ where

$$\mathcal{E}^\ell := B^{\otimes \ell} \otimes \bigoplus_{\ell(w) \leq \ell} K T_w$$

5. Schur Duality

Lem Let B be a sym alg.

If $\text{tr}(\delta(b)) = \text{tr}(b)$, $R \in (B \otimes B)^X$, $p=0$ then A is symmetric.

(Pf) Direct verification

$$\text{tr}: A \rightarrow K$$

$$b T_w \mapsto \begin{cases} \prod_i \text{tr}(b_i) & \text{if } w=1 \\ 0 & \text{otherwise} \end{cases}$$

$\leadsto$ "Schur" functor $\mathcal{F}: \mathcal{S}\text{-mod} \rightarrow eSe\text{-mod}$
 $M \mapsto eM$

Q: When can we identify $eSe \cong A$?

Lem [Curtis-Reiner 12]

If A is self-injective, T is faithful, then $\text{End}_S(T) \cong A$.

Lem [LNX'23]

If T_B contains d distinct summands $\cong B$,

(Sketch)

Consider free K -mod $\mathcal{E} = B^{\otimes d} \otimes \left(\bigoplus_{w \in \Sigma_d} K T_w \right)$ on which $B^{\otimes d}$ acts by left multn
 WTS $\mathcal{E} \cong A$ as right A -mod

Need right multn (for $a \in B^{\otimes d}$, $1 \leq i \leq d-1$)

$$f_i: b T_w \mapsto b T_w H_i$$

$$f_a: b T_w \mapsto b T_w a \quad \text{ill-defined}$$

need to fix a universal rex, and then show that such a choice doesn't matter

Grand loop argument

R[L]: f_a is ind of choice of r on $\mathcal{E}^\ell$

Q[L]: $f_i^2 = f_{s_i} f_i + f_{r_i} \in \text{End}(\mathcal{E}^\ell)^{\text{op}}$

M[L]: $f_a f_b = f_{ab}$

B[L]: $f_i f_j \dots = f_j f_i \dots$

W[L]: $f_i f_a = f_{s_i a} + f_{r_i a}$

(Pf) $R \xRightarrow{(\star 3)} W \xRightarrow{(\star 1)} M \xRightarrow{(\star 2)} Q \xRightarrow{(\star 5)} B$
 $\xleftarrow{(\star 4)}$ Thm A &

Assume that $T \in \text{mod-}A$.

$\leadsto S := \text{End}_A(T) \curvearrowright T$.

Suppose $\exists$ splitting

$$T \cong A \oplus Q \quad (\text{for some } Q \in \text{mod } A)$$

$\leadsto$ Inverse Schur functor $\mathcal{G} = \text{Hom}_A(-, T)$ takes A to T (as S -mod)

$\leadsto T \in S\text{-proj}$, $\exists e \in S$ s.t. $T \cong Se$ and $\text{End}_S(T) \cong \text{End}_S(Se) \cong eSe$

then $\exists$ canonical A -action on $T_B^{\otimes d} = T_A$ s.t.

$$T_A \cong A \otimes Q$$

Thm B holds by combining all the above.

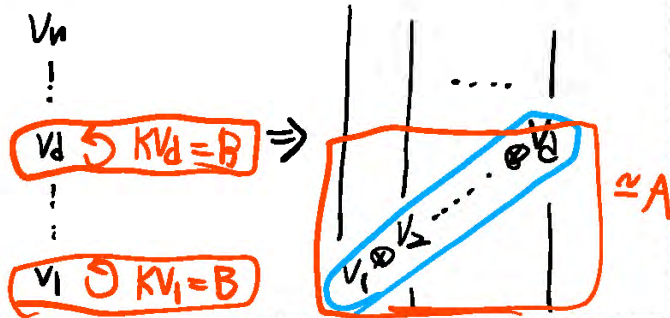
$\exists$ Splitting $\Leftarrow B^{\otimes d} | T_B \Leftarrow T_B = V(n)$ w/ $n \geq d$

$\exists$ idempotent $\Downarrow$ faithful
 $\Downarrow$ self-inj $\Leftarrow A: \text{sym} \Leftarrow B: \text{sym}$
 $\exists$ Schur functor

eg. $\mathcal{H}_q(\Sigma_d) = K\mathcal{H}_d$, $B=K$

$T_B = V(n)$

$T_A = V(n)^{\otimes d}$



Cor If $n \geq d$, then Schur duality for $\mathcal{H}_q(\Sigma_m)$ holds

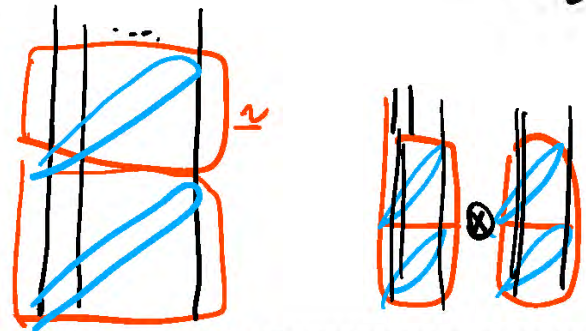
Cor [Euseev-Kleshchev'17] If $n \geq d$, then Schur duality for $B^{\otimes d} \rtimes K\Sigma_d$ holds

$\leadsto$ appln in proving Turner's conjecture

eg. $\mathcal{A}(m) = \mathcal{H}_q(\Sigma_m) \rtimes \mathcal{H}_2$, $B = \mathcal{H}_q(\Sigma_m)$

$T_B = V(n)^{\otimes m}$

$T_A = T_B \otimes T_B$



Cor If $n \geq 2m$, then Schur duality for $\mathcal{A}(m)$ holds

(for $\mathbb{Q}\Sigma_m \rtimes \Sigma_2$ holds)

Rmk Schur dualities for AHA, AK alg have similar pictures.

However, our argument doesn't go through $\nsubseteq B = K[X^{\pm 1}]$ is not Frobenius.

6. Further Connections

① $AHA \simeq K^{GL_d \times \mathbb{C}^*}(\mathbb{Z}_d)$

$\leadsto A \simeq K^?(\mathbb{Z}_?)$

② $\mathbb{Z}(AHA) \xrightarrow{\sim} eAHAe$: Satake isom
 $f \mapsto fe$

$\leadsto \mathbb{Z}(A) \xrightarrow{\sim} eAe$. What's e ?

③ $(B^{\otimes d})^{\Sigma_d}$ has basis $\{p_\lambda\}$ $\leftarrow m$ polyn
 $\frac{m \text{ id}}{\langle p_\lambda, p_\lambda \rangle} = \pi \frac{\sim}{\sim}$

$\leadsto m \text{ id for } \mathcal{A}(m)?$

④ $NH_n \hookrightarrow \text{HOM space for KLR alg}$
 $\uparrow U^*(\mathbb{Z})$

$\text{GWP} \hookrightarrow \text{HOM space for gKLR alg}$
 $\uparrow \text{Nichols alg}$